



M002 Week 4 — English Worksheet

Topic: AND and OR Conditions — Smarter Decisions · **Course:** Maze Madness · **Time:** about 45 minutes

Last week your agent used **AND** (both true). This week you add **OR** (**either** one true) and use them **together** in the same maze.

- **AND** → the action happens **only when both** conditions are true.
- **OR** → the action happens **when at least one** condition is true.

1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
if wall ahead AND wall on right:
    turn left
```

Both must be true. The agent has a wall ahead but an open path on the right — does it turn left?


```
if no wall ahead OR no wall below:
    move forward
```

Only one needs to be true. The path ahead is blocked but the path below is open — does the agent move?


```
keep doing forever:
  if wall ahead AND wall on right:
    turn left
  otherwise if no wall ahead OR no wall on left:
    move forward
  otherwise:
    turn right
```

The agent reaches a corner with walls on the front AND right. What does it do? At an open junction, which branch runs?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The agent should turn left **only when** there is a wall ahead AND a wall on the right (a dead-end corner). Right now it turns at *any* wall.

```
if wall ahead OR wall on right:  
    turn left
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The agent should move forward when **either** the path ahead is clear OR the path below is clear.

```
if no wall ahead AND no wall below:  
    move forward
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — At a fork the agent should **turn right if it can** (path on the right), otherwise turn left. Right now the order is wrong, so it always turns left first.

```
if no wall on left:
    turn left
otherwise if no wall on right:
    turn right
```

Hint: check the right side first.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Solve the maze using **both** an **AND** condition and an **OR** condition. When you finish, come back here.

Send a photo or video of the agent reaching the end, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

I used **AND** when ...

I used **OR** when ...

One tricky moment was when ...

To fix it, I ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while the agent runs the maze. Talk through it like you are teaching someone who has never seen it. Try to use these words: **AND**, **OR**, **condition**, **both**, **either**.

1. Show the start of the maze, then run your code.
2. Read each `if ... AND ...` and `if ... OR ...` block out loud and say what makes it run.
3. Show one bug you hit and how you fixed it.
4. Say in your own words how **AND** is different from **OR**.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.